

Intake Manifold

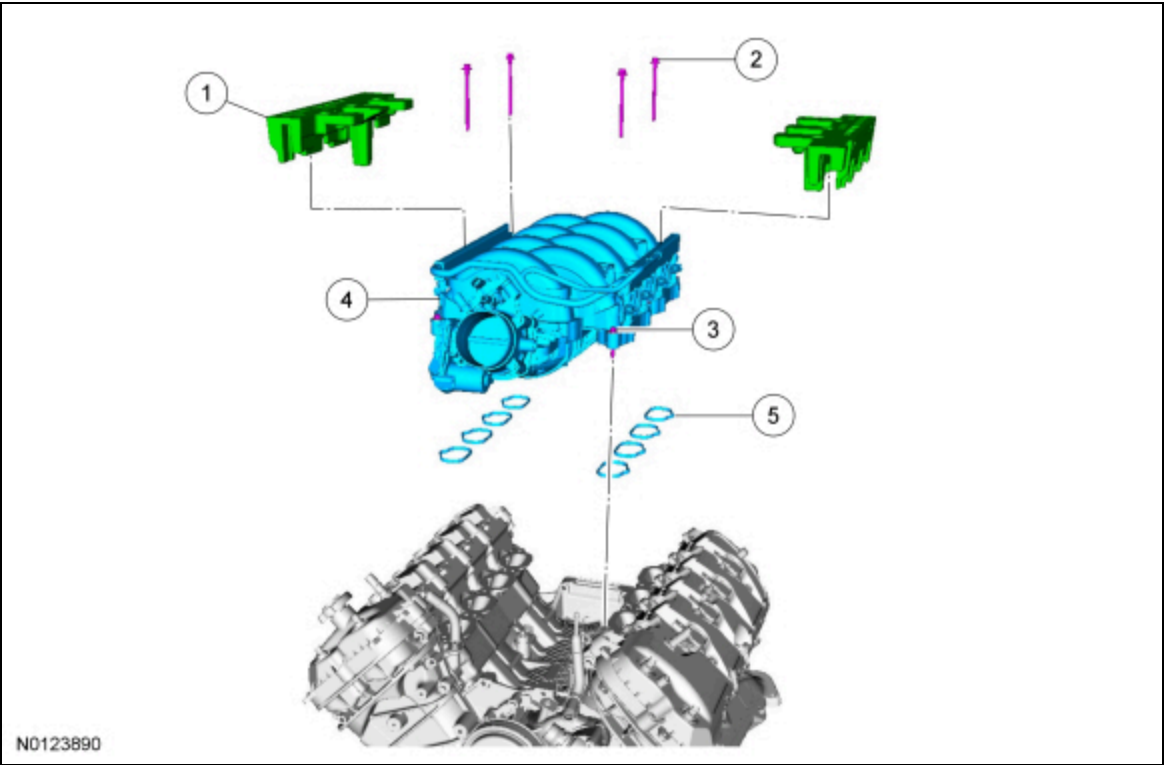


Material

Item	Specification
Motorcraft® Metal Surface Prep Wipes ZC-31-B	—

NOTE: [Refer to the on-line Workshop Manual to view this illustration as an interactive exploded view, requires Adobe® Acrobat® 8.0 or higher.](#)

NOTE: [Refer to the on-line Workshop Manual to learn about using an Interactive Illustration.](#)



Item	Part Number	Description
1	6P0136P013	Fuel rail insulator (2 required)
2	W714864W714864	Fuel rail bolt (4 required)
3	9Y4509Y450	Intake manifold bolt (part of 9424) (6 required)
4	94249424	Intake manifold
5	94399439	Intake manifold gasket

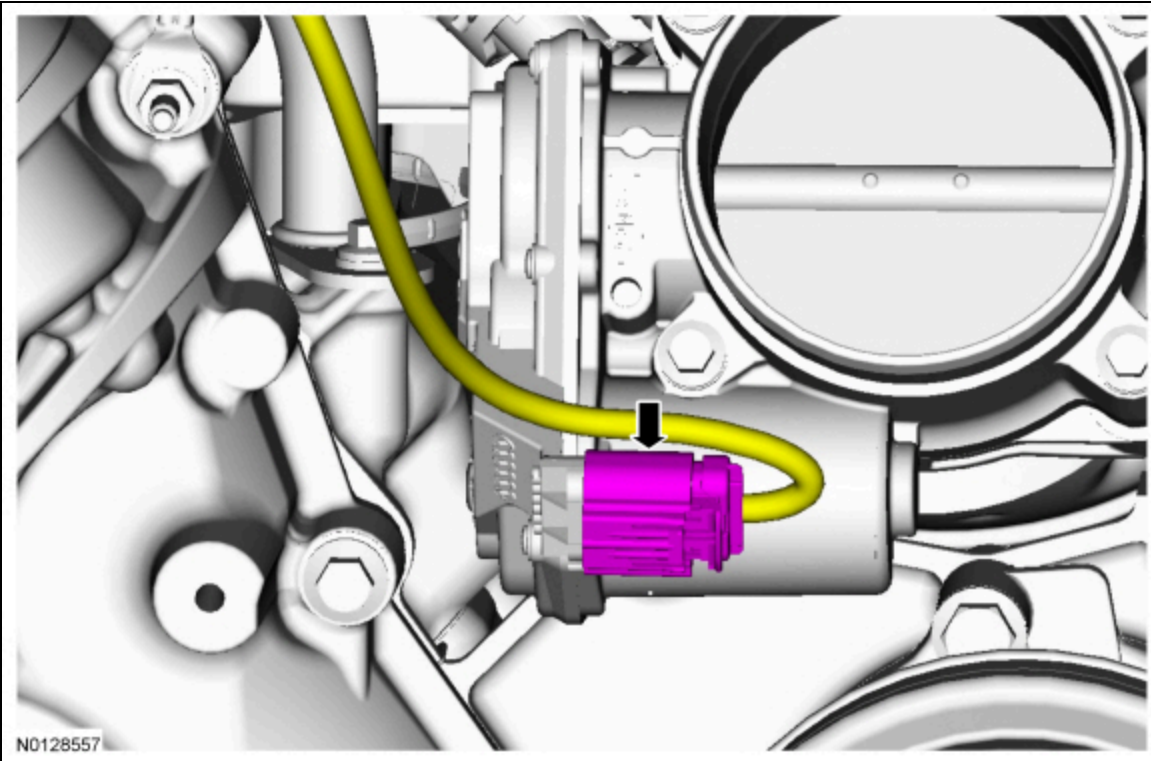
Removal

WARNING: Do not smoke, carry lighted tobacco or have an open flame of any type when working on or near any fuel-related component. Highly flammable mixtures are always present and may be ignited. Failure to follow these

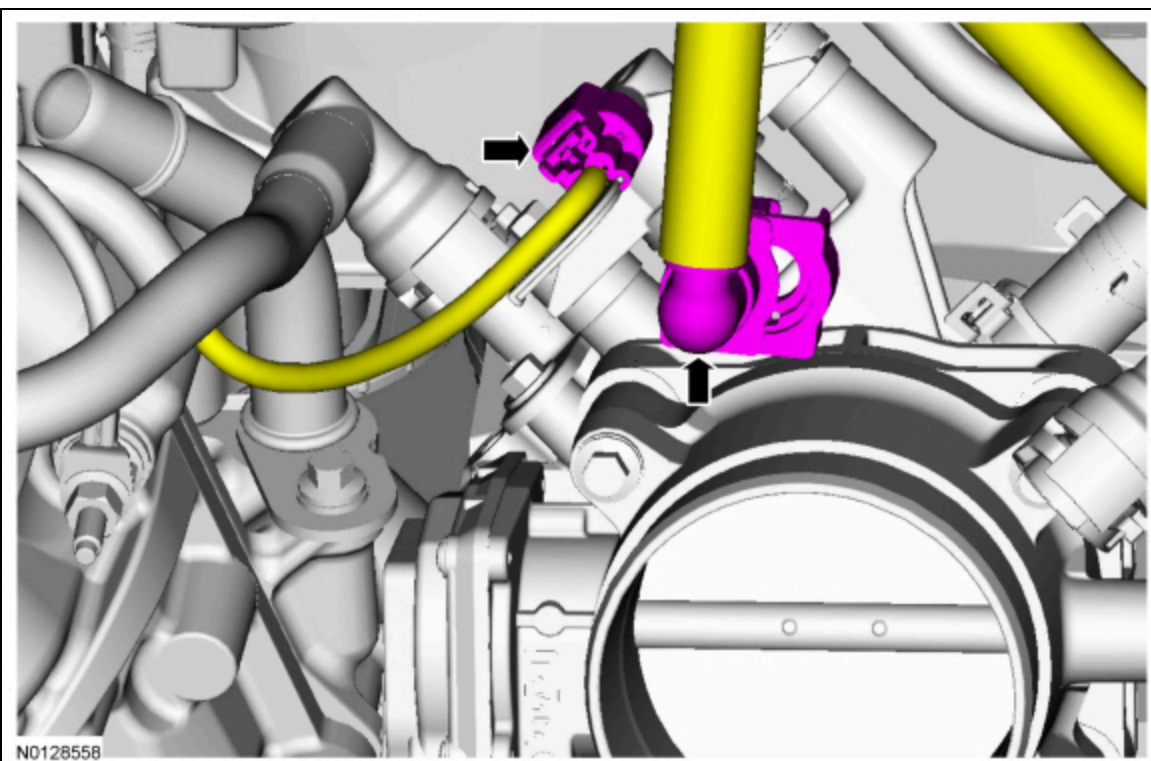
instructions may result in serious personal injury.

WARNING: Before working on or disconnecting any of the fuel tubes or fuel system components, relieve the fuel system pressure to prevent accidental spraying of fuel. Fuel in the fuel system remains under high pressure, even when the engine is not running. Failure to follow this instruction may result in serious personal injury.

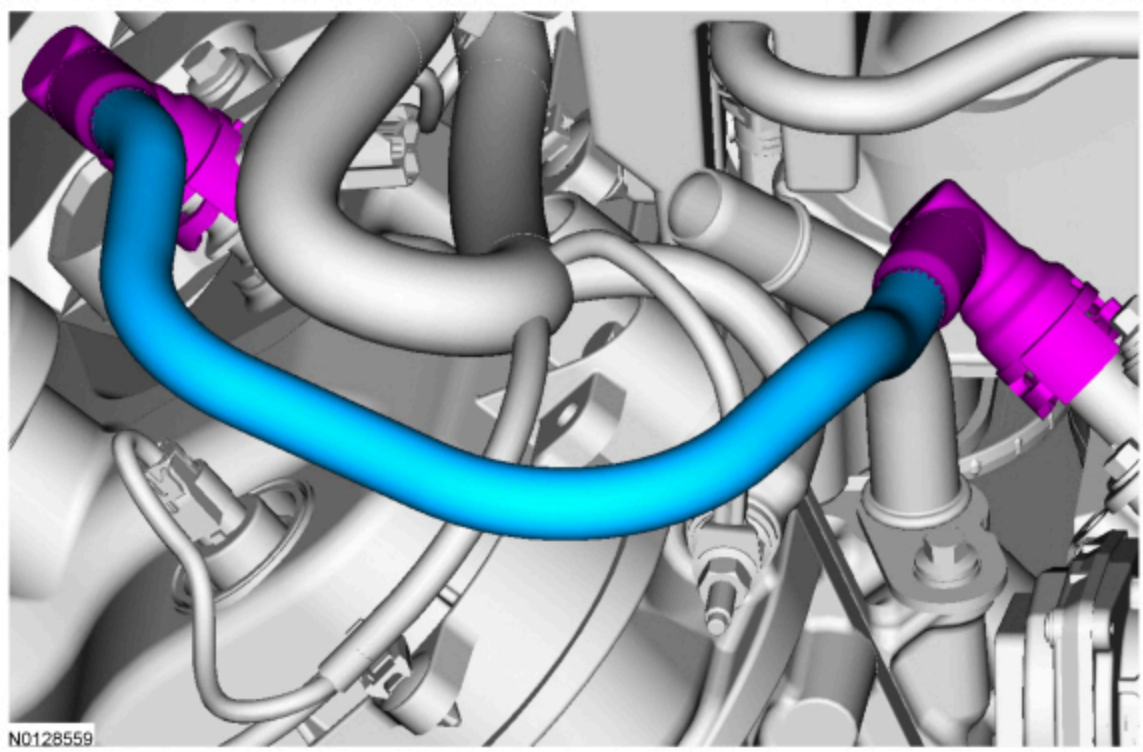
1. Release the fuel system pressure. For additional information, refer to Section 310-00.
2. Remove the Air Cleaner (ACL) outlet pipe. For additional information, refer to Section 303-12.
3. Disconnect the Throttle Body (TB) electrical connector.



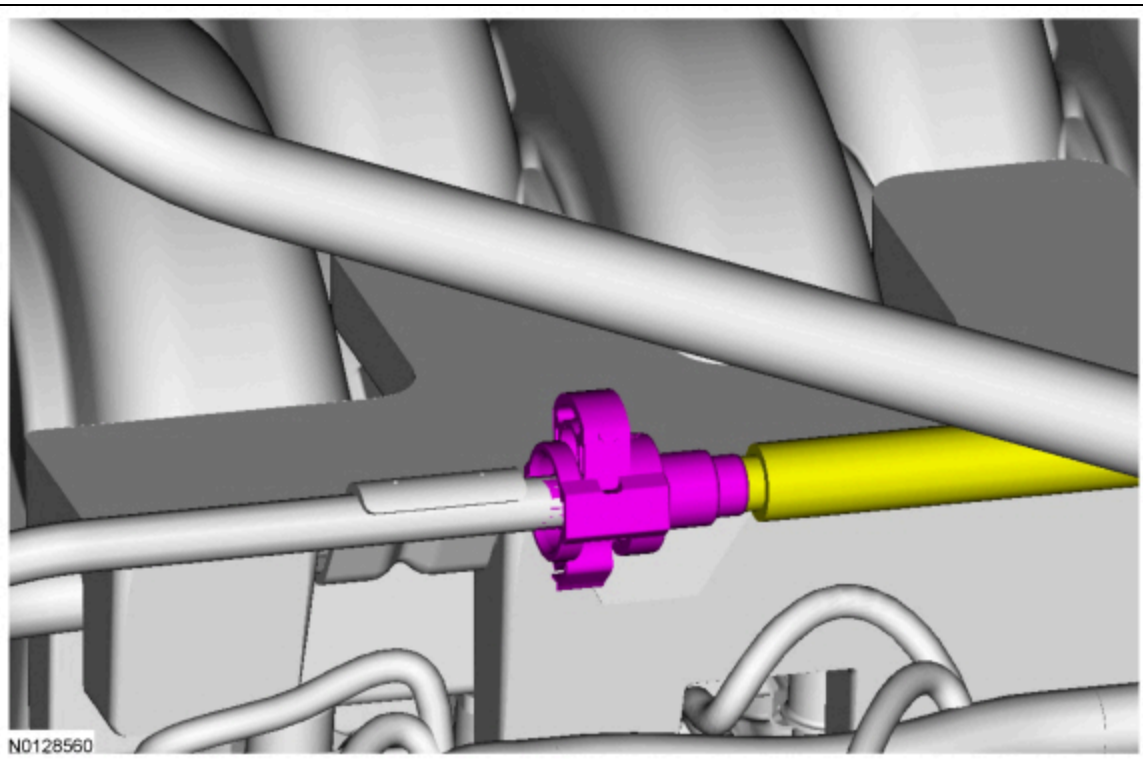
4. Disconnect the Evaporative Emission (EVAP) canister purge valve tube and electrical connector. For additional information, refer to the quick connect coupling procedure in Section 310-00.
 - Detach the EVAP tube retainer from the fuel rail.



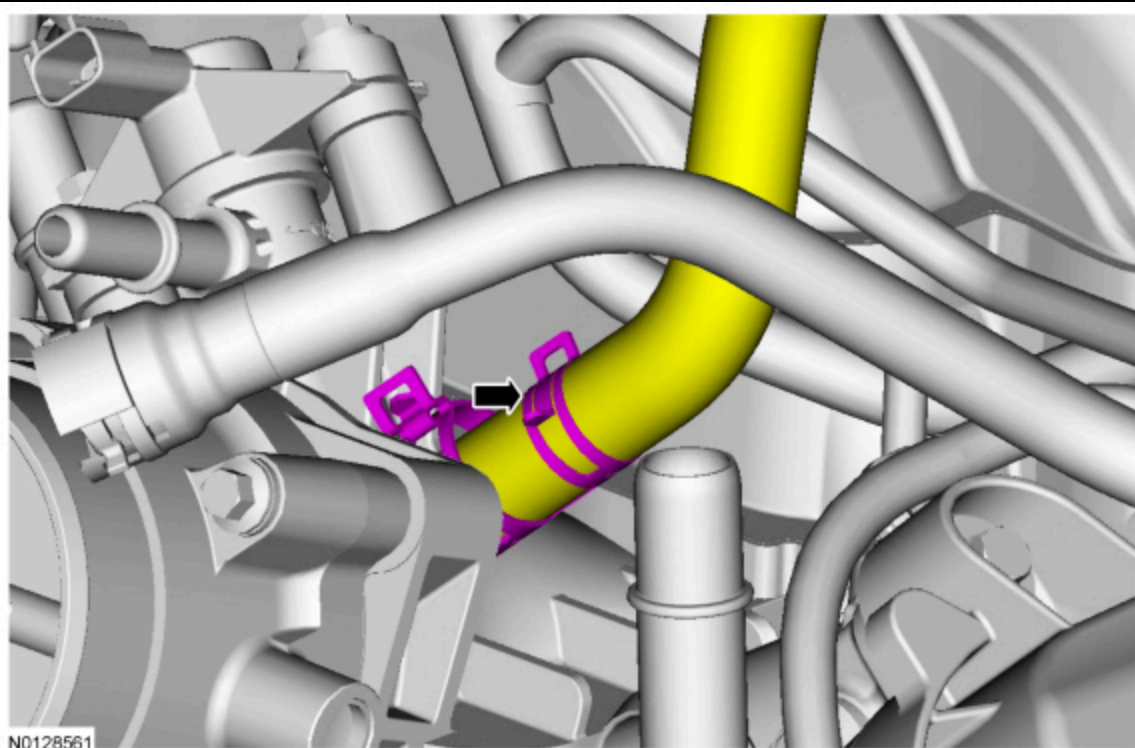
5. Remove the crankcase ventilation tube. For additional information, refer to the quick connect coupling procedure in Section 310-00.



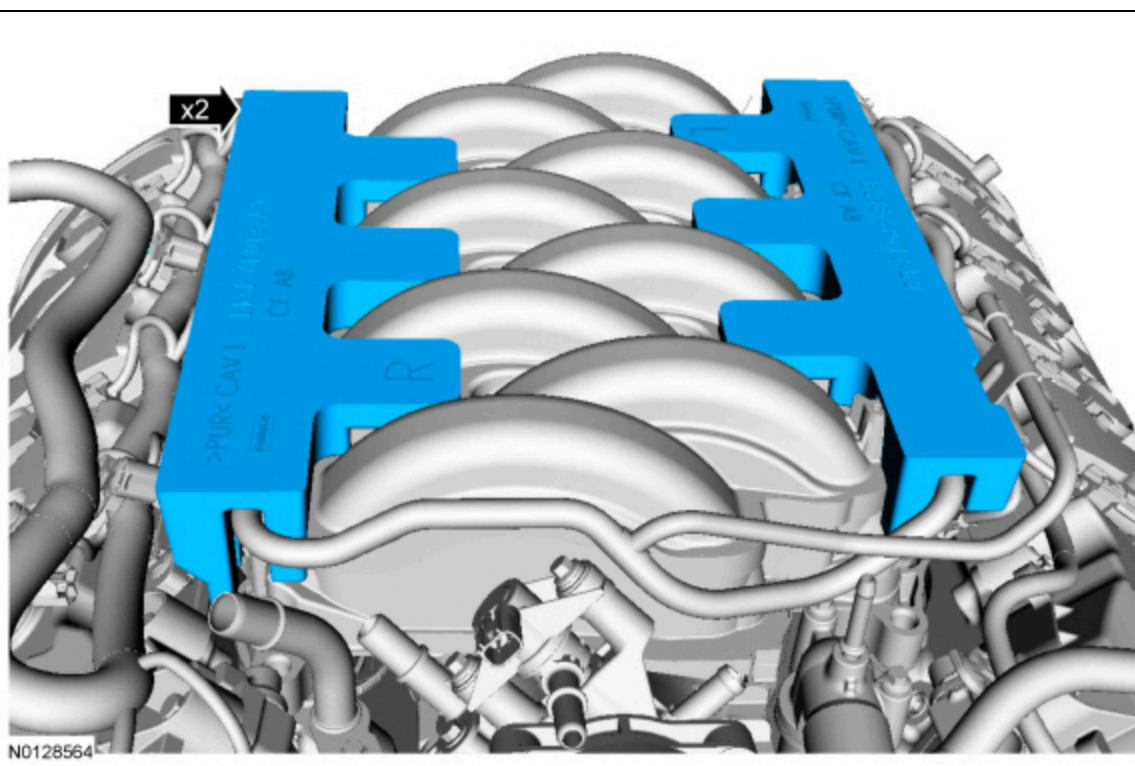
6. Disconnect the fuel supply tube. For additional information refer to the quick connect procedure in Section 310-00.



7. Release the clamp and disconnect the vacuum tube from the intake manifold.

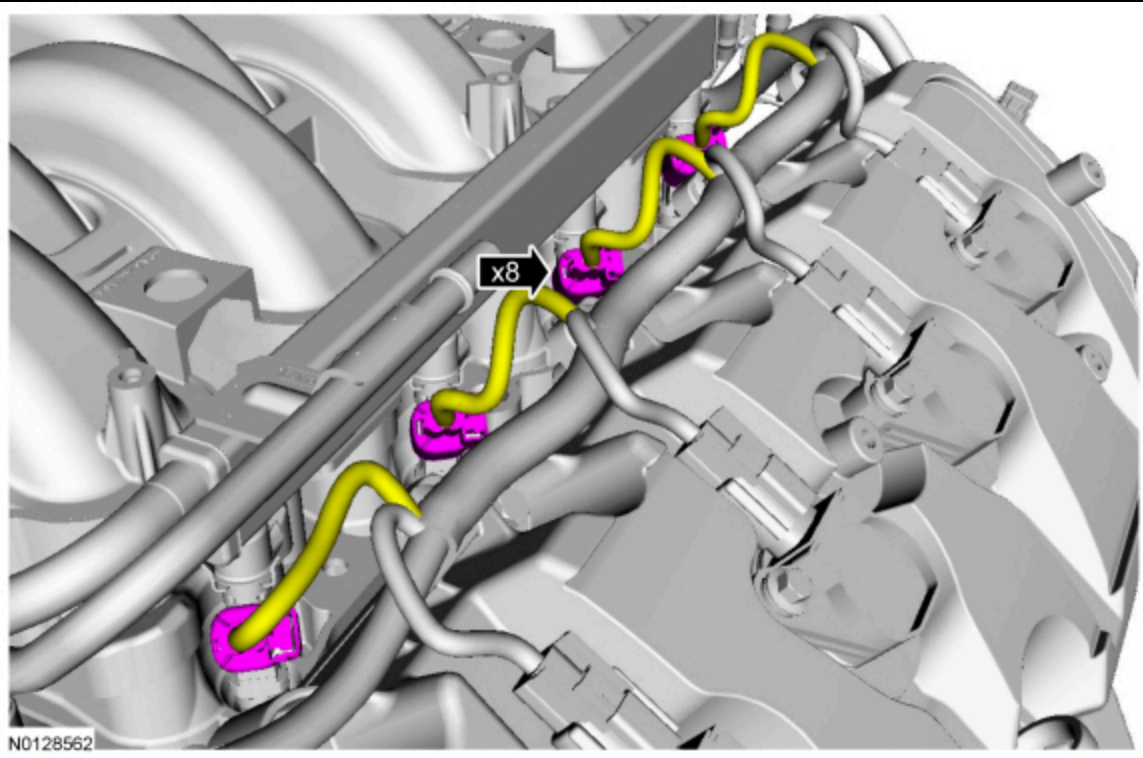


8. Remove the LH and RH fuel rail insulators.

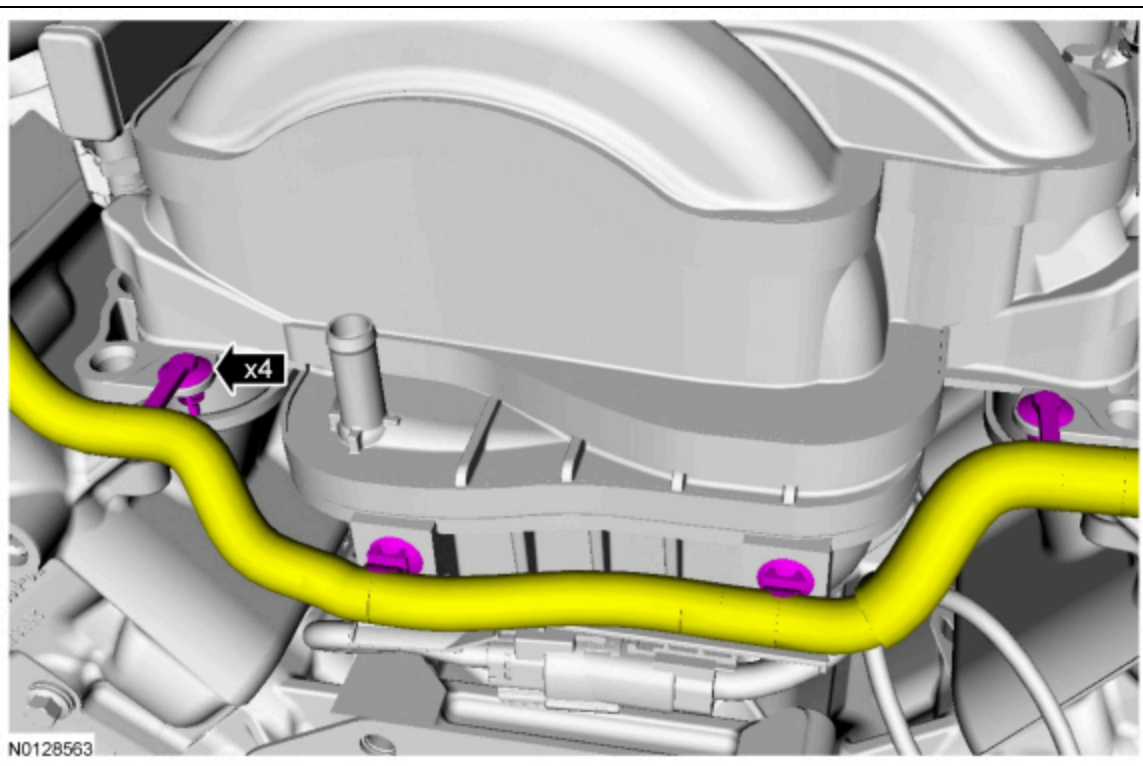


9. **NOTE:** The 4 LH fuel injector electrical connectors are shown, the 4 RH fuel injector electrical connectors are similar.

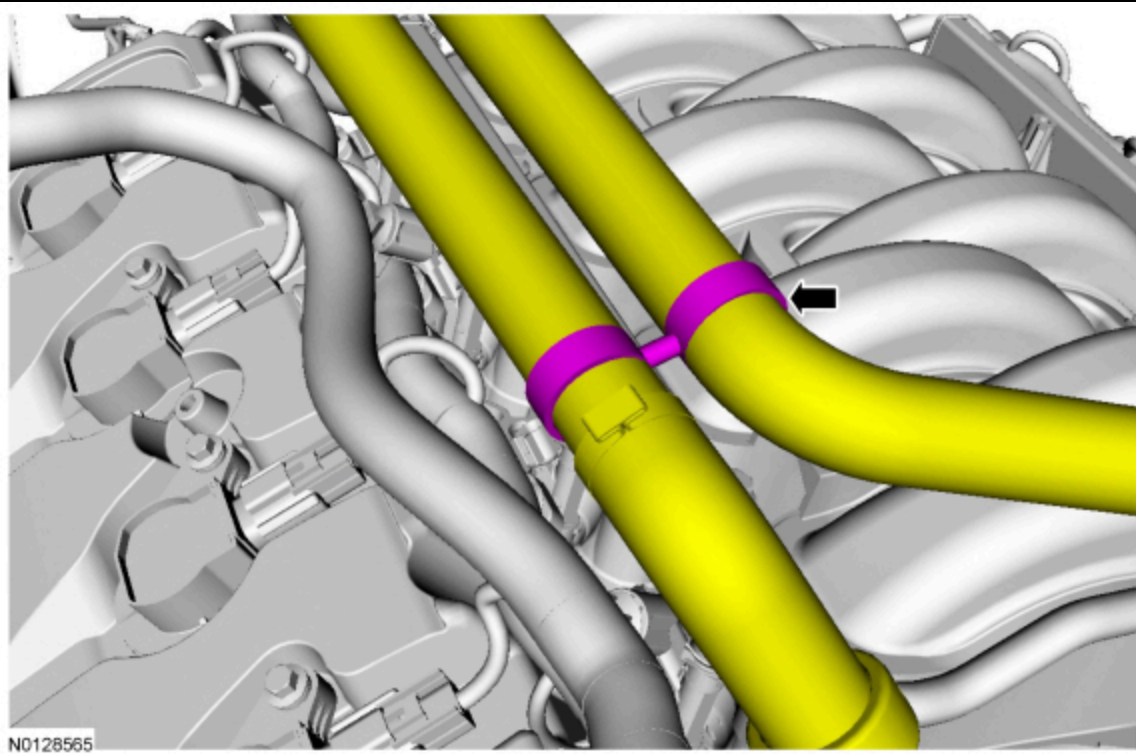
Disconnect the 8 fuel injector electrical connectors.



10. Detach the 4 wiring harness retainers from the rear of the intake manifold.

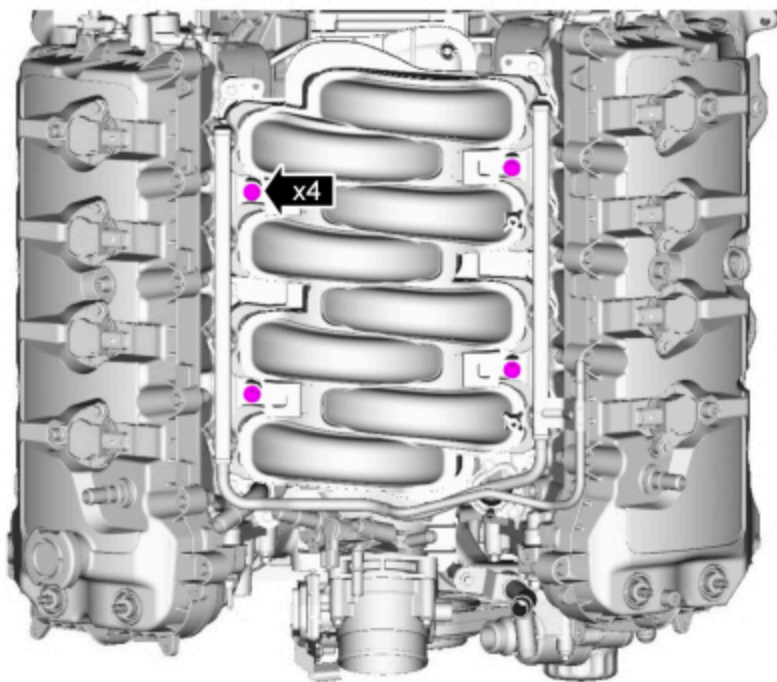


11. Detach the heater hose retainer and position the heater hose aside.



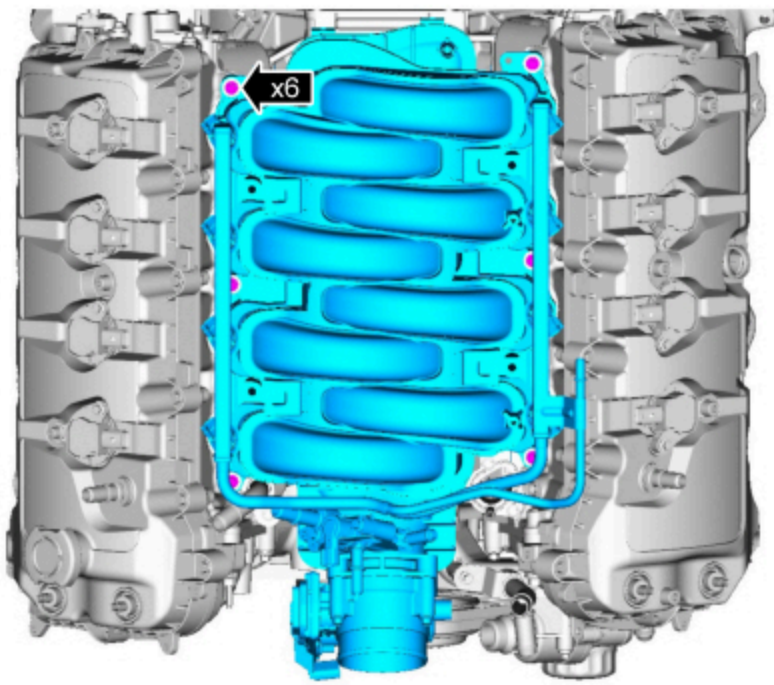
12. **NOTE:** *It is not necessary to remove the fuel rail from the intake manifold assembly.*

Remove the 4 fuel rail bolts.



13. Loosen the 6 bolts and remove the intake manifold.

- Discard the gaskets.



N0128566

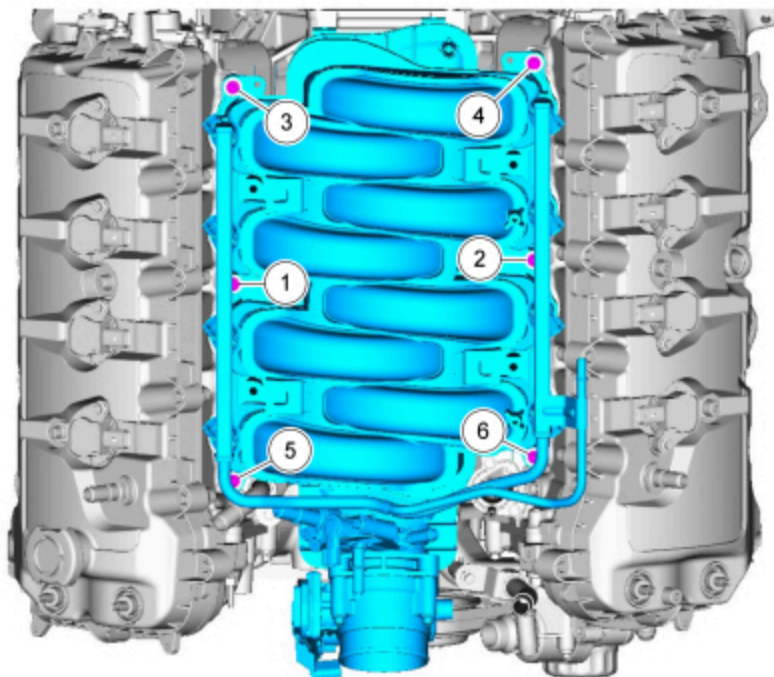
Installation

1. **NOTICE:** If the engine is repaired or replaced because of upper engine failure, typically including valve or piston damage, check the intake manifold for metal debris. If metal debris is found, install a new intake manifold. Failure to follow these instructions can result in engine damage.

NOTE: Clean the sealing surfaces with metal surface prep. Follow the directions on the packaging. Inspect the mating surfaces.

Using new gaskets, install the intake manifold.

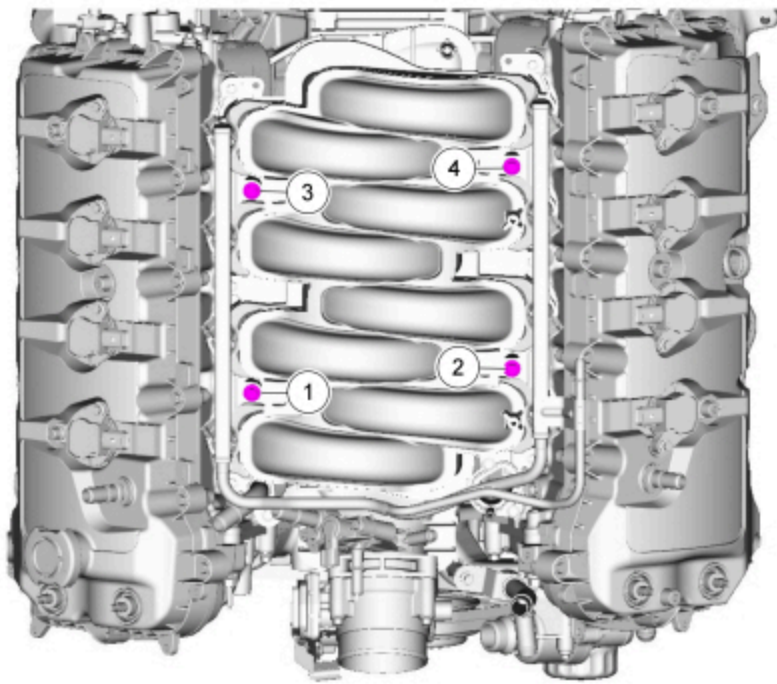
- Tighten the 6 intake manifold bolts in the sequence shown in 2 stages.
 - Stage 1: Tighten to 10 Nm (89 lb-in).
 - Stage 2: Tighten an additional 45 degrees.



N0123891

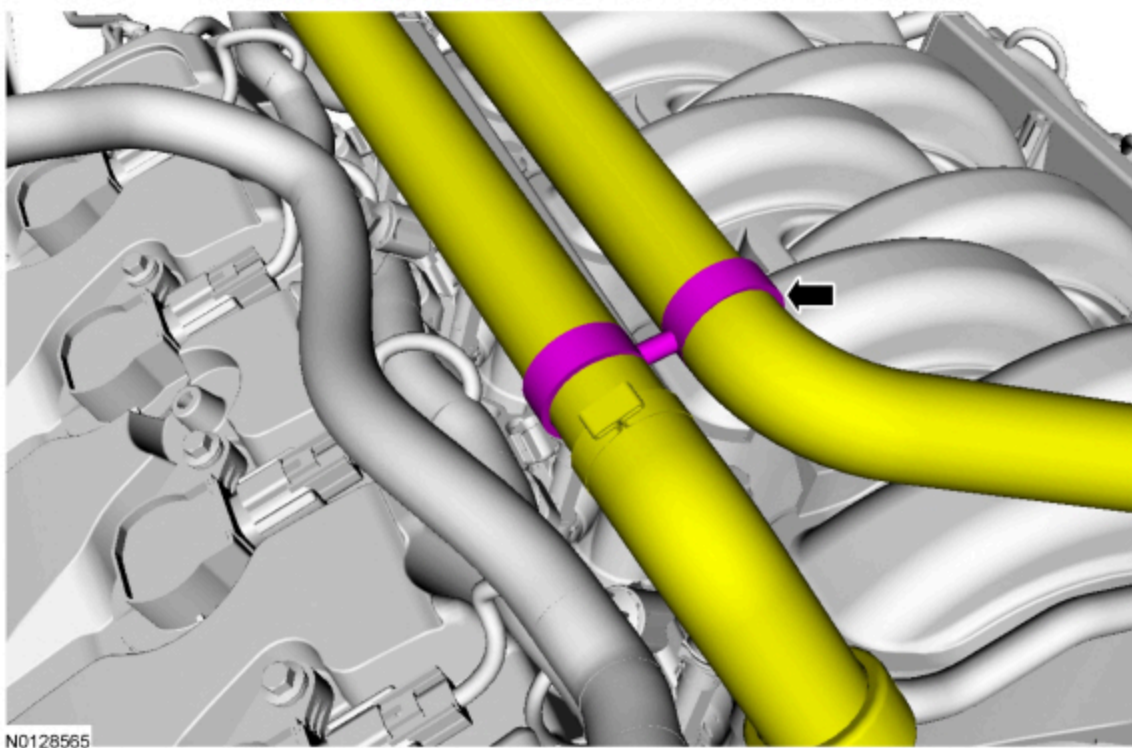
2. Install the 4 fuel rail bolts.

- Tighten in the sequence shown in 3 stages.
 - Stage 1: Hand tighten.
 - Stage 2: Tighten to 10 Nm (89 lb-in).
 - Stage 2: Tighten an additional 90 degrees.



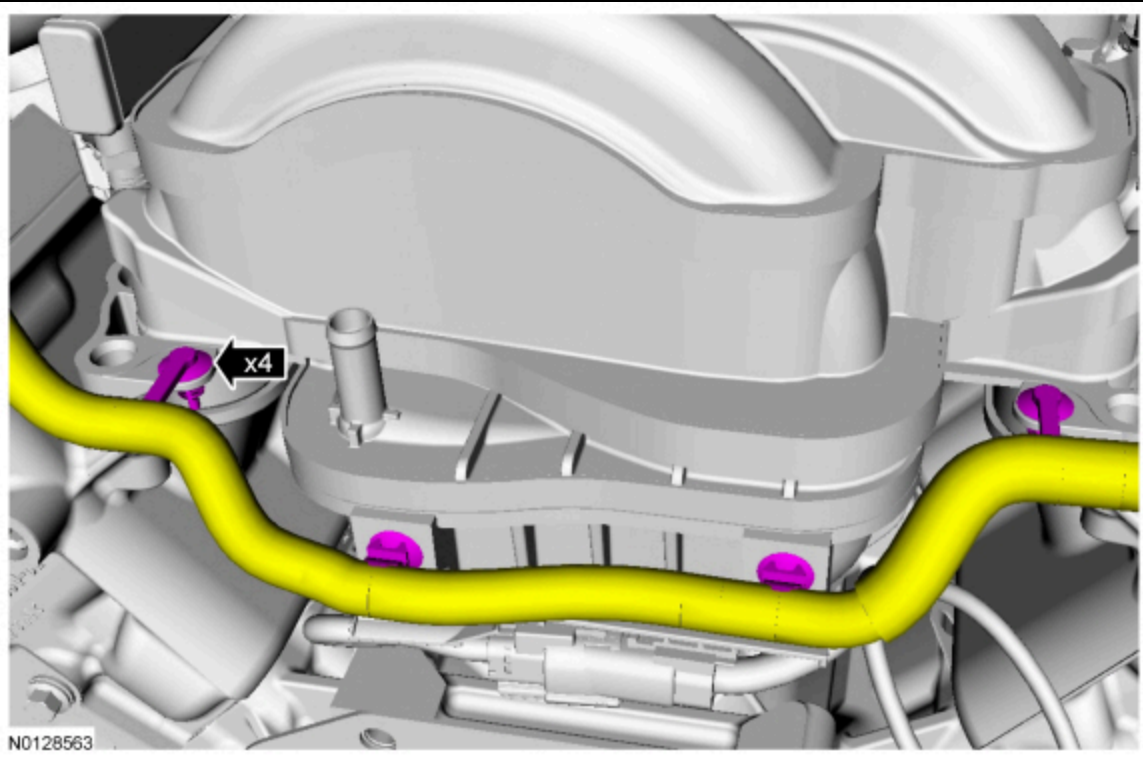
N0123892

3. Position the heater hoses and attach the heater hose retainer.



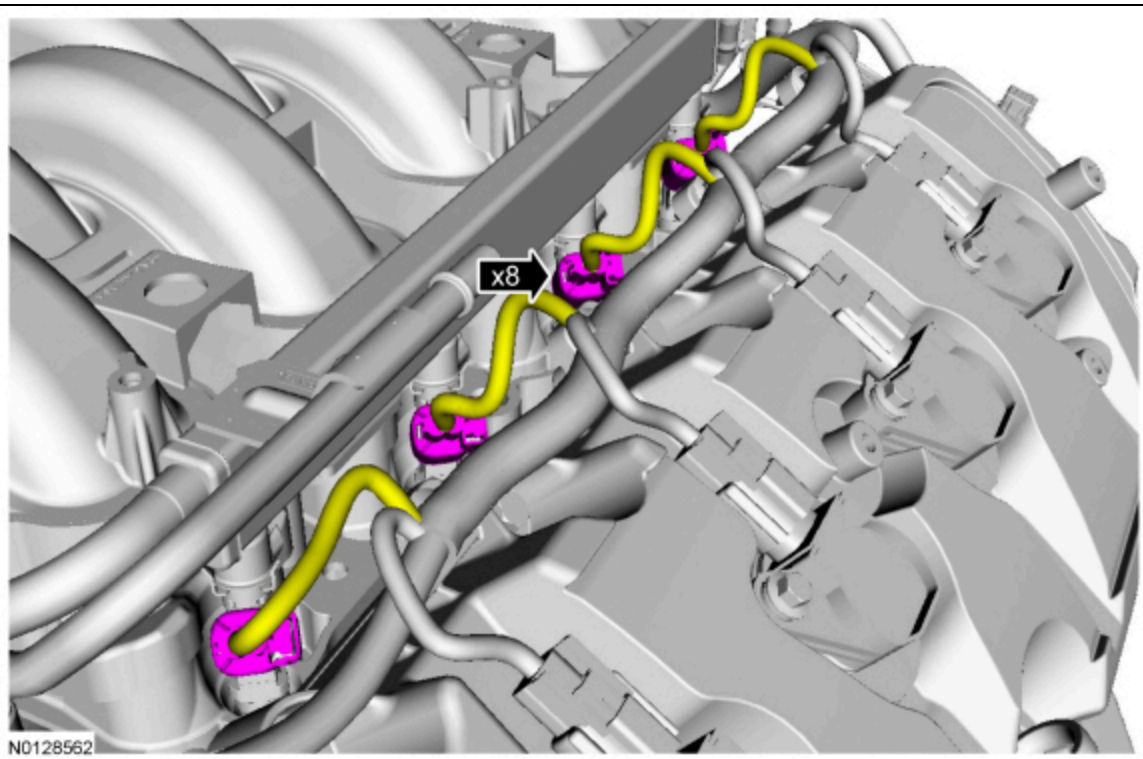
N0128565

4. Attach the 4 wiring harness retainers to the rear of the intake manifold.

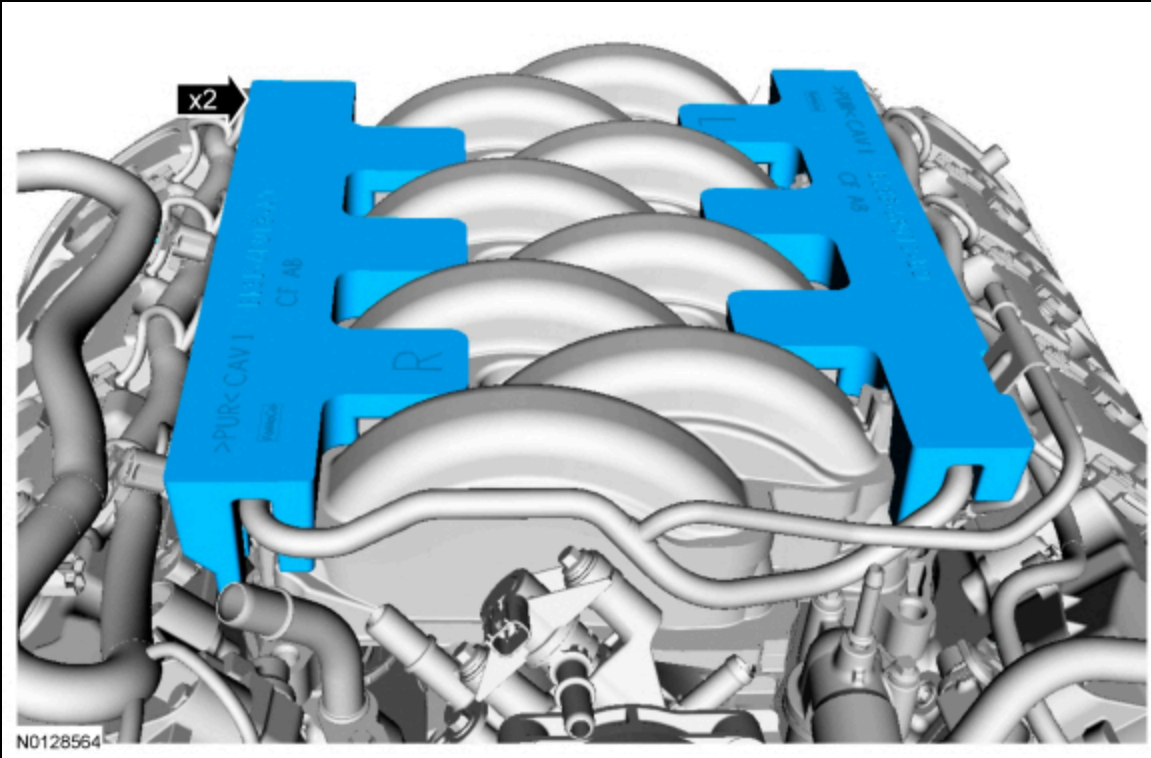


5. **NOTE:** *The 4 LH fuel injector electrical connectors are shown, the 4 RH fuel injector electrical connectors are similar.*

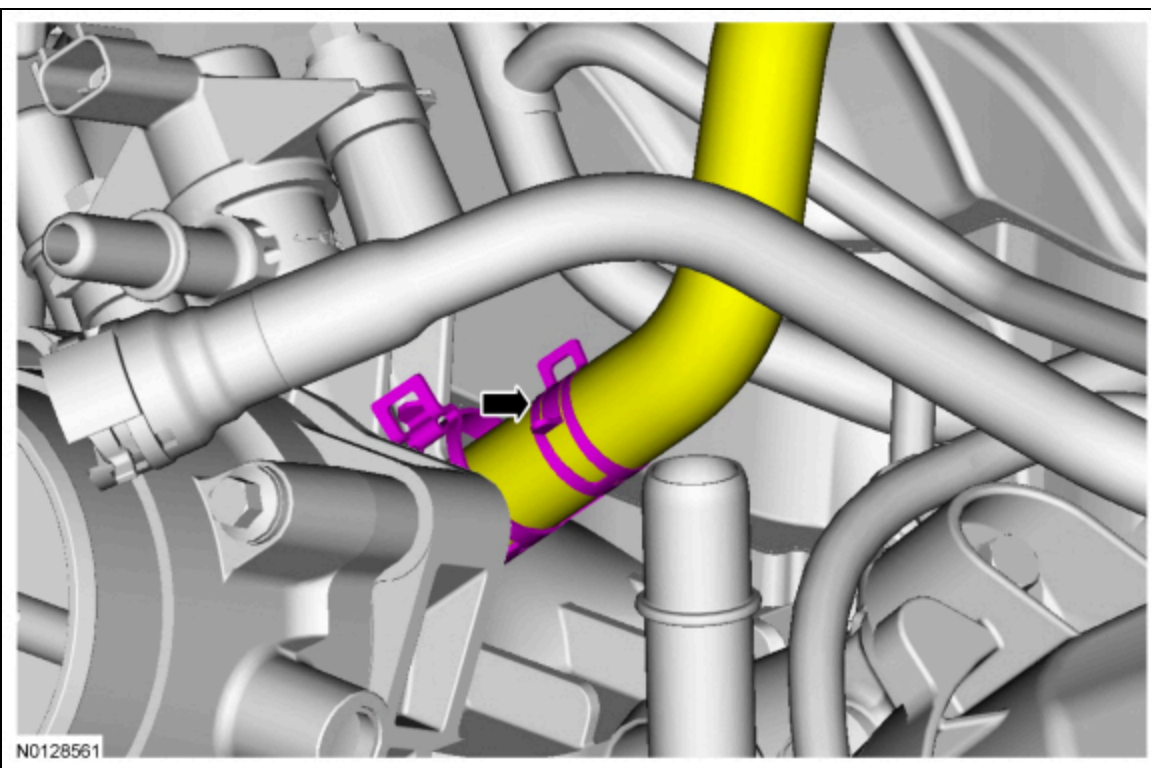
Connect the 8 fuel injector electrical connectors.



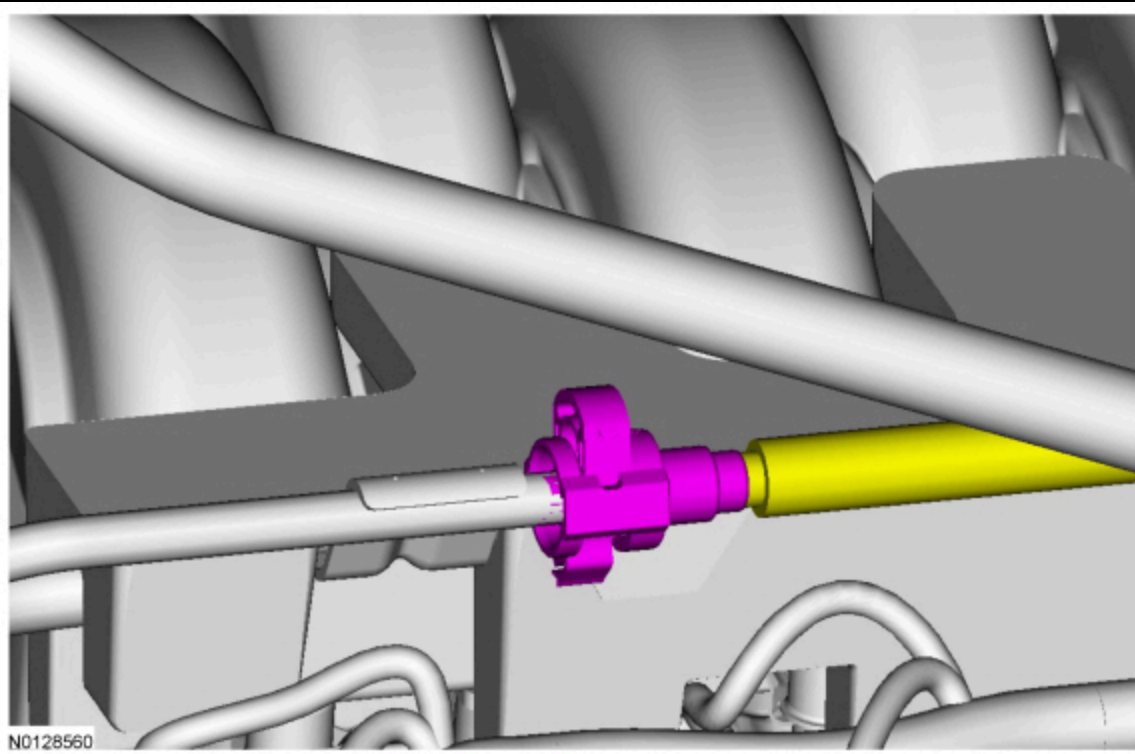
6. Install the LH and RH fuel rail insulators.



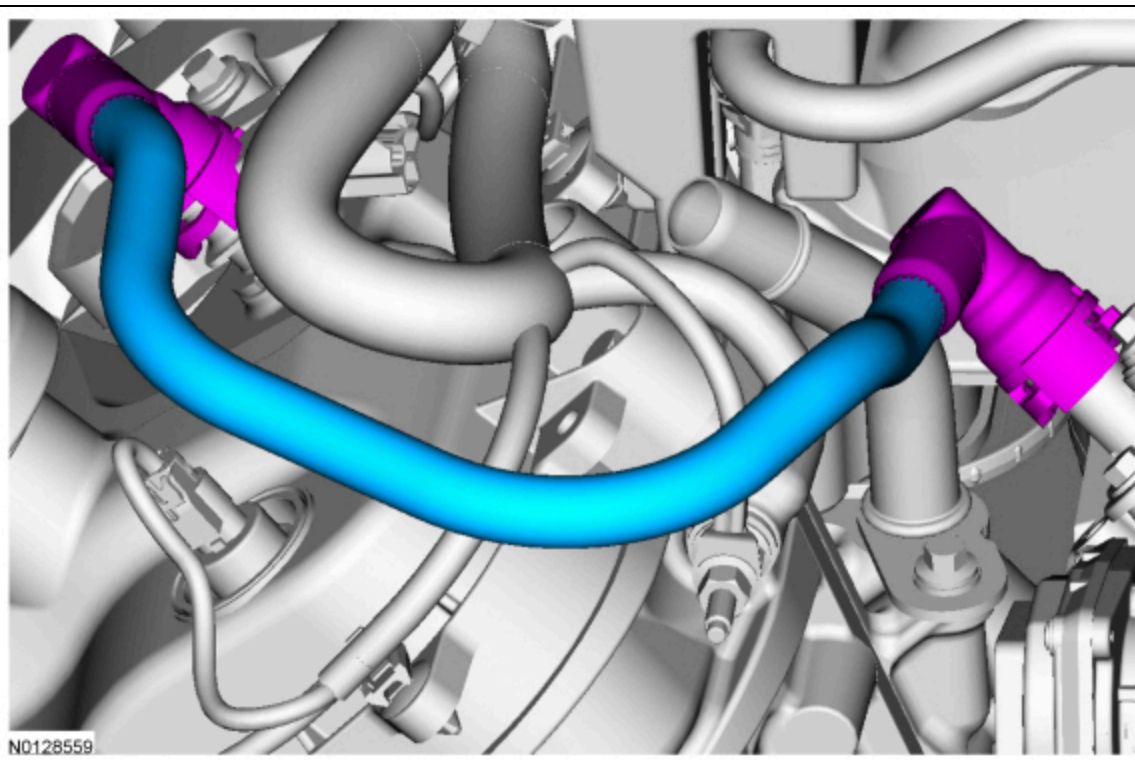
7. Connect the vacuum tube to the intake manifold.



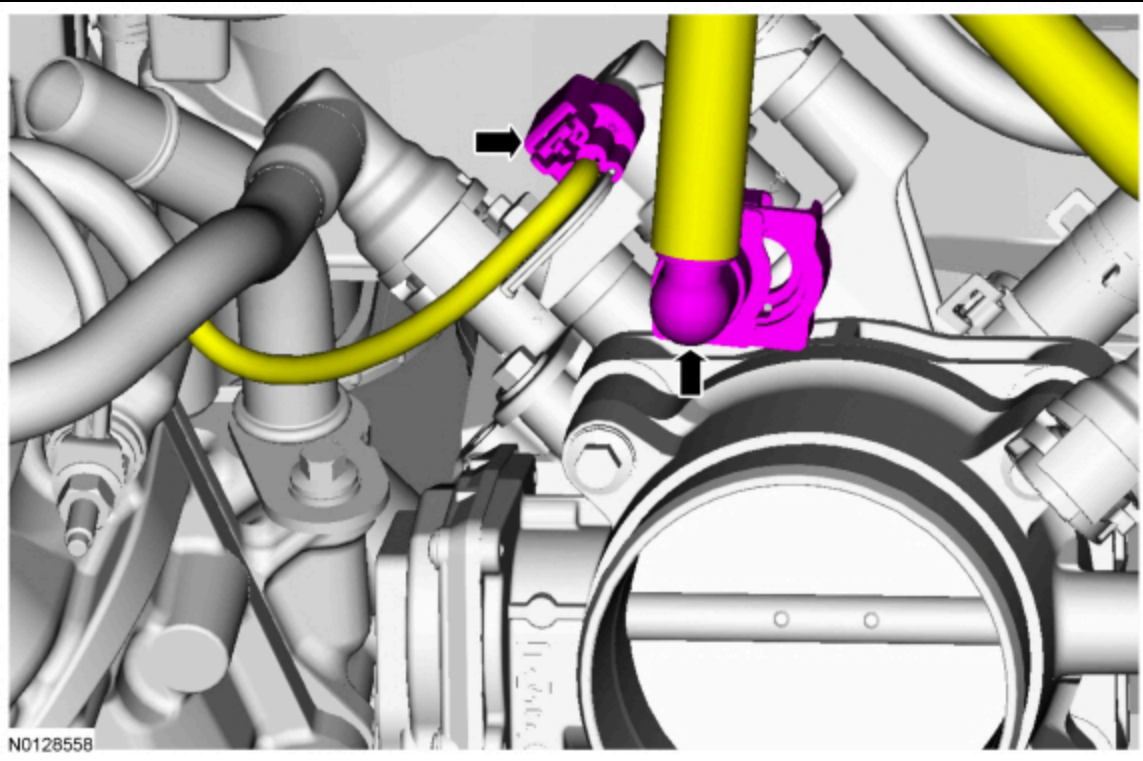
8. Connect the fuel supply tube. For additional information, refer to the quick connect coupling procedure in Section 310-00.



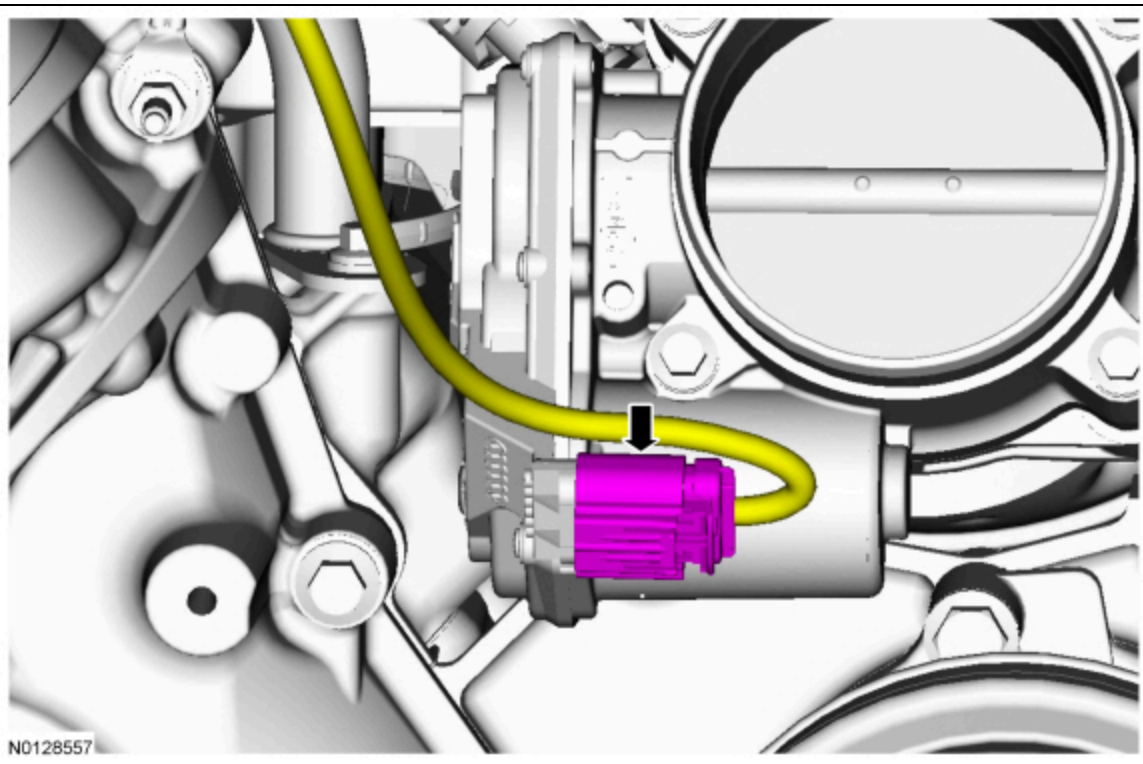
9. Install the crankcase ventilation tube. For additional information, refer to the quick connect coupling procedure in Section 310-00.



10. Connect the Evaporative Emission (EVAP) purge valve tube and electrical connector. For additional information, refer to the quick connect coupling procedure in Section 310-00.
- Attach the EVAP tube retainer to the fuel rail.



11. Connect the TB electrical connector.



12. Install the Air Cleaner (ACL) outlet pipe. For additional information, refer to Section 303-12.